

DashAR: An Implementation of Augmented Reality Technology for Automotive Applications

A Thesis Defense Presentation by Trevor D. Brown

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Presentation Flow

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History

XR, Automobile Instrumentation, and the Heads Up Display.

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Opportunity

A new approach to Automotive HUDs.

03

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Investigations into Automotive HUD R&D.

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What can become the modern Automotive HUD?

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Designing an AR-based Automotive HUD System.

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Developing the DashAR System.

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Where can it go next?

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Thanks

Expression of my gratitude.

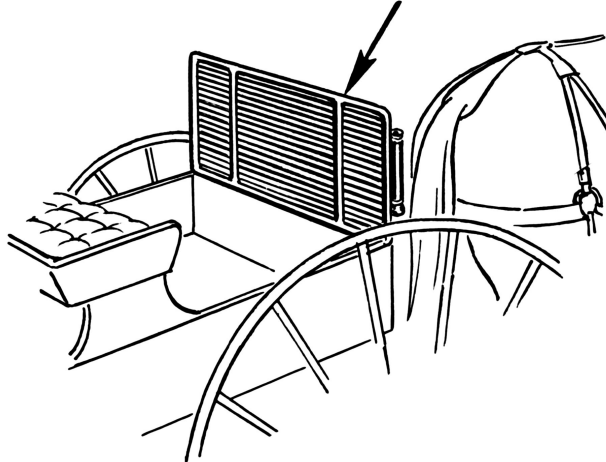
01

History

XR, Automobile Instrumentation, and the Heads Up Display.

History of Automotive Instrumentation

DASHBOARD



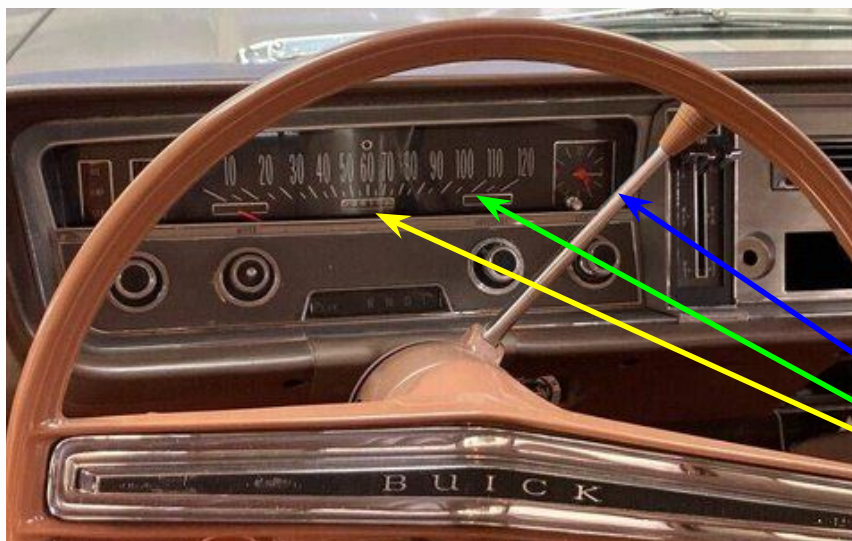
Etymology

(19th Century)
Horse-drawn Carriages



Ammeter

(1925)
Ford Model T



Evolving Dash

(1964)
Buick Skylark

Clock
Speedometer
Gear Shift Pos. Ind.



Modern Dashboard

(2017)

Subaru Forester



OBDII

(1996)

History of HUDs



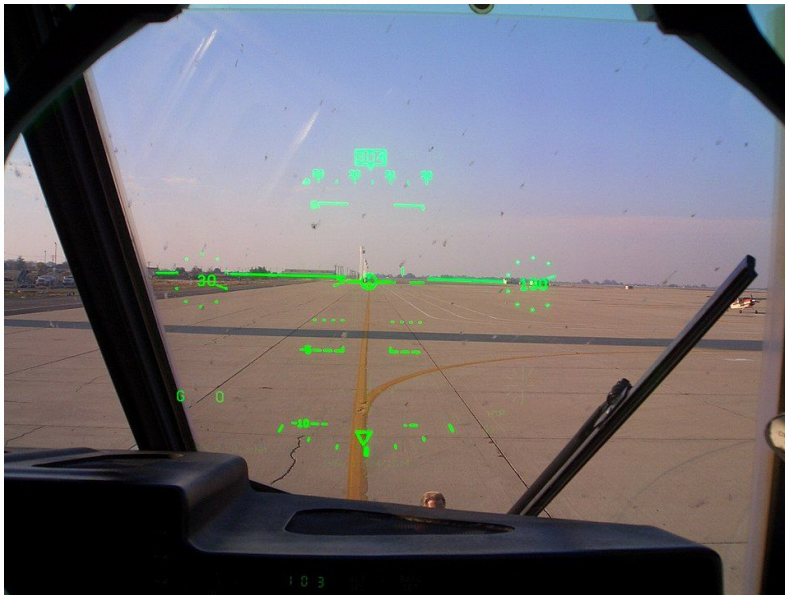
Third-Party HUDs

(2025)



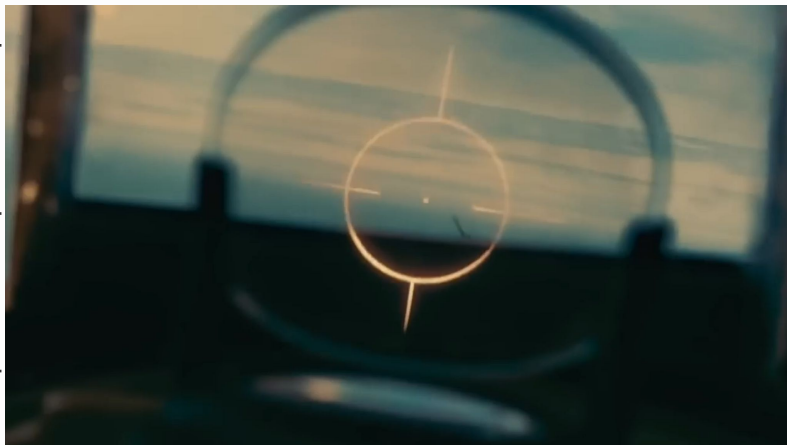
Automobile HUDs

(1980 - present)
BMW E60



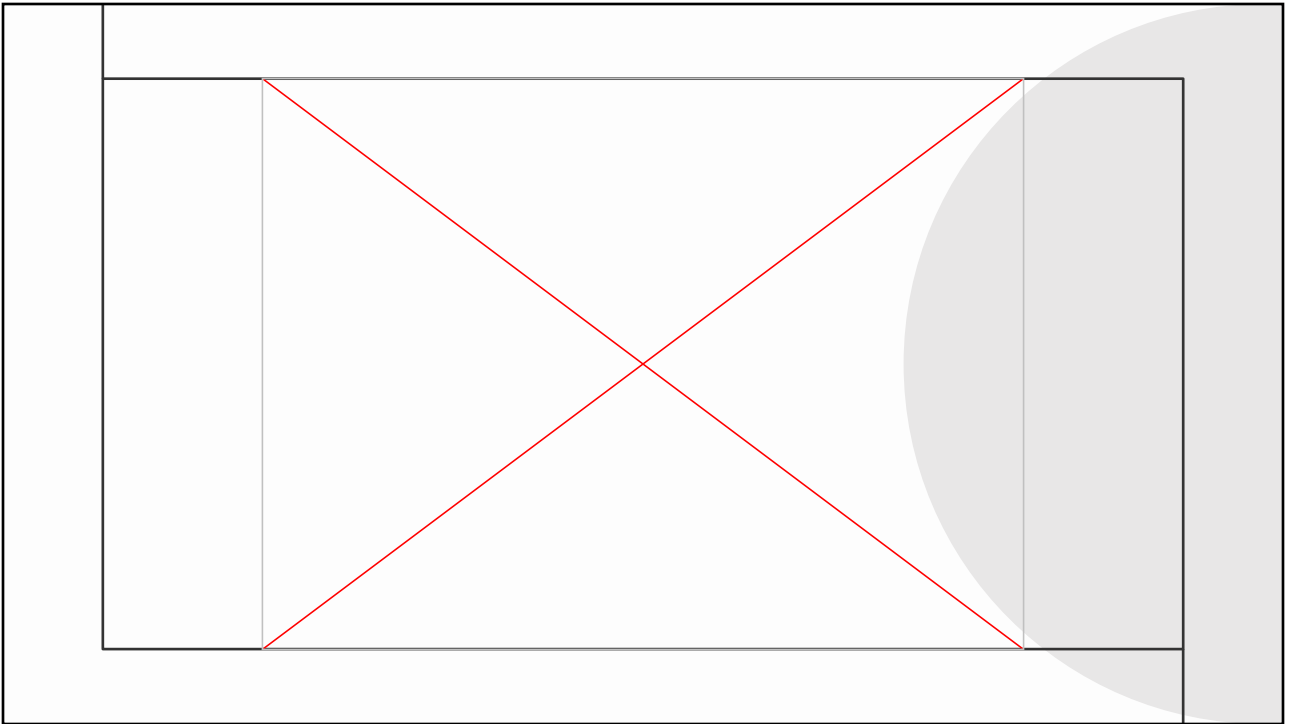
Modern Fighter Jet HUD

(1999)
Lockheed Martin C-130J
Super Hercules



Fighter Plane HUDs

(1942 - present)
Frame from *Dunkirk* (2017)



History of Extended Reality (XR)



Stereopsis

1838, by Charles Wheatstone



Sensorama

1956, by Morton Heilig



The Sword of Damocles

1968, by Ivan Sutherland





**Meta Quest
3S**
(2024)

02

Opportunity

A new approach to Automotive HUDs.

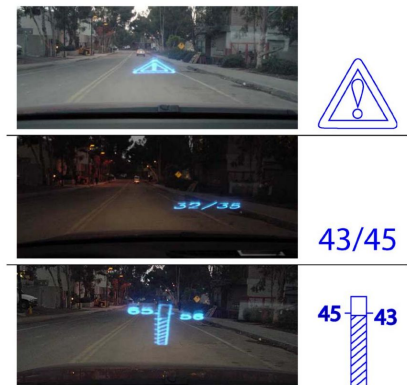
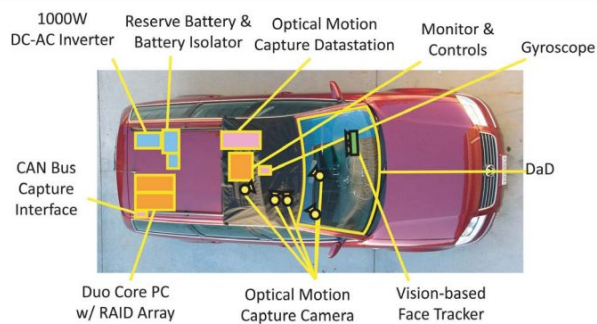


03

Prior Research

Investigations into Automotive HUD R&D.

Literature Review



A. Doshi, S. Yuanhsien, et. al. (2009). *A Novel Active Heads-Up Display for Driver Assistance*. IEEE Transactions on Systems, Man, and Cybernetics - Part B.

- Windshield Implementation of a HUD while driving; different alert styles for speed limit violations.

Literature Review (continued)

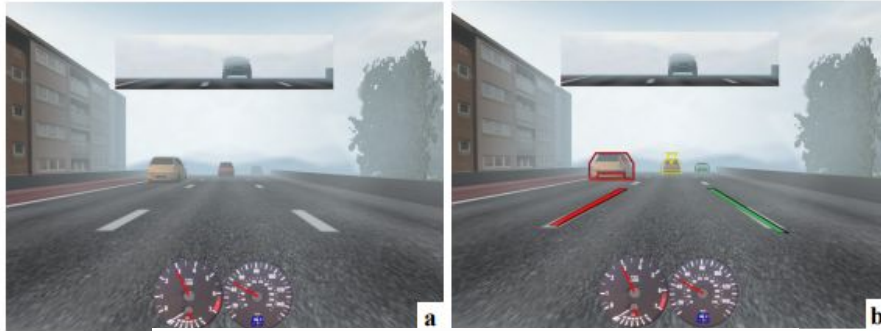
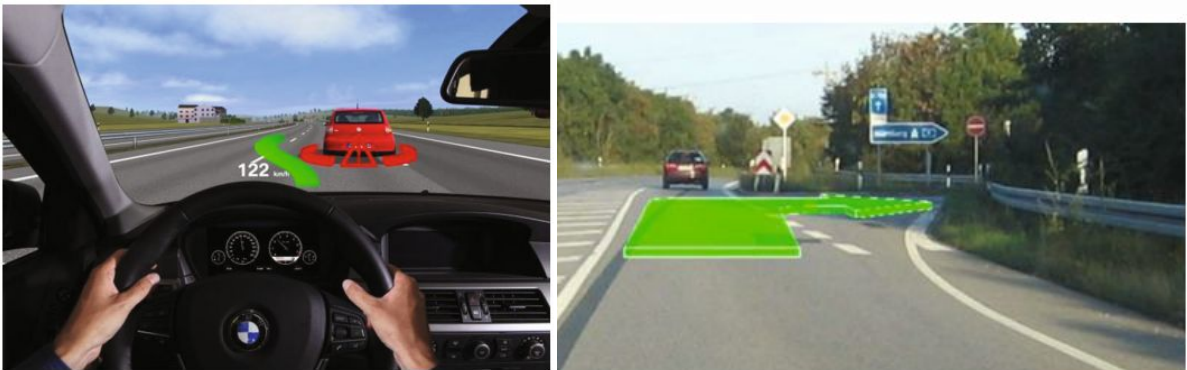


Figure 1: Driving under low visibility conditions
a) With contemporary HDD, and b) With proposed HUD visual aid

V. Charissis and S. Papanastasiou. (2006). *Human-Machine Collaboration Through Vehicle Head Up Display Interface*. European Annual Conference on Human Decision-Making and Manual Control.

- Heads Up vs. Heads Down Display. Consideration for low visibility conditions.

Literature Review (continued)



- A. Pauzie. (2015). *Head Up Display in Automotive: A New Reality for the Driver*. International Conference on Human-Computer Interaction.
 - Reasoning for improved “eyes on the road” time using HUD.

04

Project Objectives

What can become the modern Automotive HUD?

The DashAR System

Why is it named DashAR?

DashAR is not just an app... it's a system.

DashAR's Features

Works with most Automobiles

User-customizable HUD

Data Aggregation

Real-Time Delivery

Platform-Agnostic

DashAR's Source Code

Open Sourced by Design

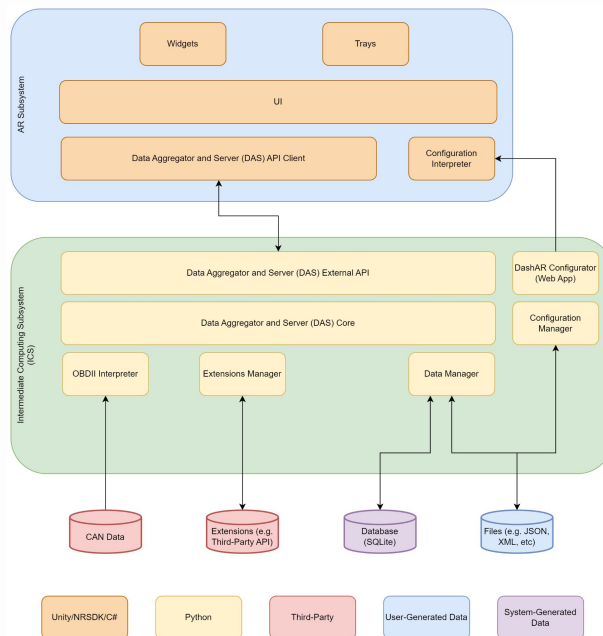
Distributed via GitHub

<https://github.com/TrevorDBrown/DashAR> - The DashAR System

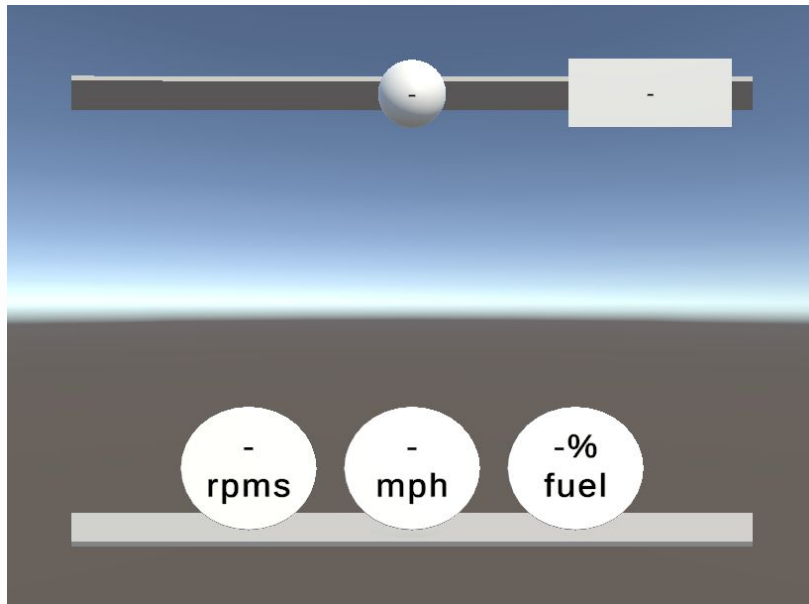
<https://github.com/TrevorDBrown/DashAR-Docs> - DashAR's Documentation

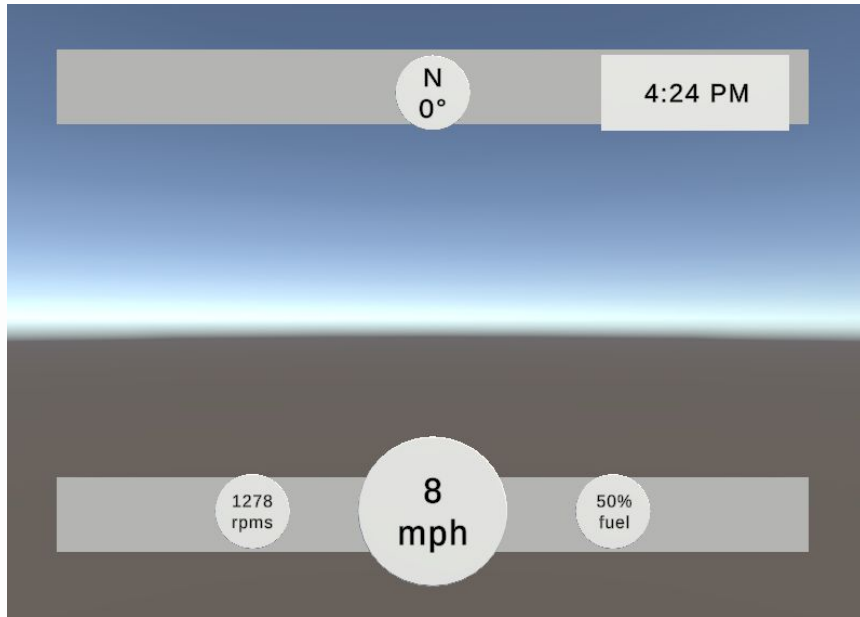
Design

Designing an AR-based Automotive HUD System.



The Ideal Target AR Platform





06

Timeline

Planning for the future.

Project Schedule

Task	Details	Jun 2024	Jul	Aug	Sep	Oct	Nov	Dec	Jan 2025	Feb	Mar	Apr
Foundations	Design, procuring devices.											
Develop	DAS, HUD											
Test	Integration and Systems Tests											
Defense	Submission and Defense of Thesis.											

Completed

Partially Completed

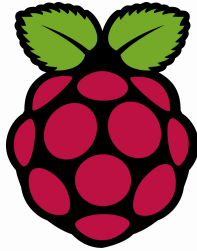
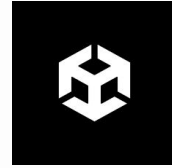
In Progress

07

Implementation

Developing the DashAR System.

Platforms and Tools



Cost to Implement DashAR

Intermediate Computing Subsystem

Raspberry Pi 5 (Computer and Power Supply) - \$92

12V to 120V Power Inverter (150W) - \$40

Micro SD Card (64 GB) - \$13

Surge Protector - \$10

OBDII-to-Serial - \$35

AR Subsystem

XREAL Air 2 Pro - \$400

XREAL Beam Pro - \$200

Total Cost

\$ 790

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End Results

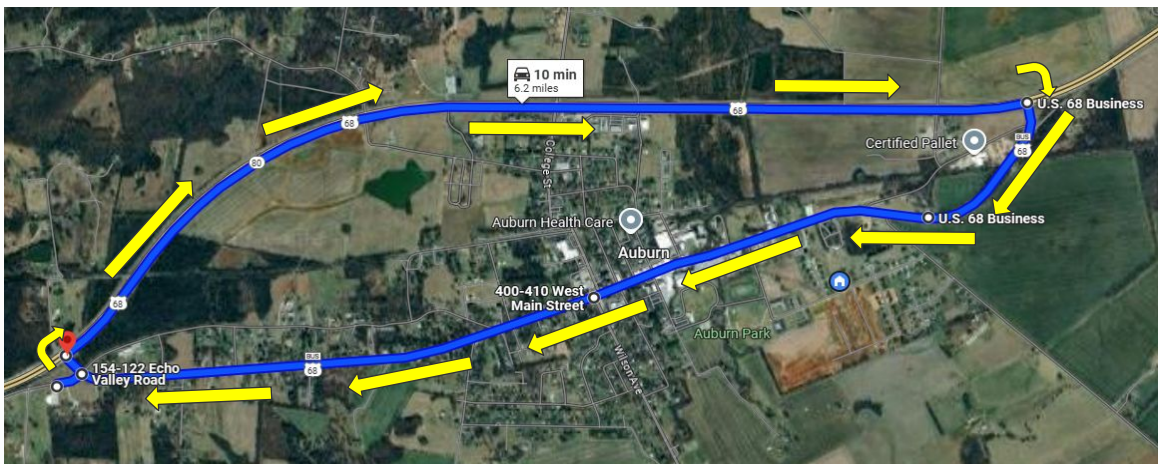
DashAR in Action!

Test Drive (11/10/2024)





Test Drive (11/10/2024)





09

Dilemmas

Development hurdles and problems, and the race against time.

Time Constraints

Data Aggregator and Server

Extensions not fully implemented
HUD Customization App not implemented

HUD

Unable to develop “aesthetically pleasing” UI
Customizations must be handled through JSON configurations.

UX

Unable to thoroughly test with subjects, although a UX test plan was developed.

ICS/DAS API Technical Challenges

Hardware

Safe power down (e.g. UPS, battery pack)
Establishing internet connectivity (e.g. cellular modem, hotspot)

Data Sourcing

Data Sources must be directly configured in the API (i.e. no “codeless” data sourcing)

Connectivity Issues

Some inconsistency in connectivity when starting the ICS.

HUD Technical Challenges

Optimized Data Request Methodology

Client/Server (synchronous) vs. Publisher/Subscriber (asynchronous)

Enforcing “Distraction Free” Operation

Restrictions to additional apps while driving.

User Experience Concerns

Usable in any conditions (rain, snow, night, etc.)

HUD Self-Imposed Constraints

Tray Anchors

Widgets must be placed in specific positions along the trays.

Five tray anchors per tray, totaling 10 tray anchors.

Widget Sizes

Three widget sizes: Circle (Small), Circle (Large), and Rectangle (Large)

Future Works

What's next for DashAR? Where can it go next?

Planes, Trains, and Automobiles!

Operational Data

Instrumentation, Operating Conditions, Weather, Hauled Load Data (for trucks), Customized Navigation (for military applications)

Intelligent Operation

Computer Vision applications (Blind Spot Visibility, Object Detection), Operating Limit Monitoring and Alerts (e.g. unsafe vehicle acceleration, excess load weight when hauling, etc.)

Luxurious Extensions

Infotainment Integration, Climate Control, Multi-view Perspectives via camera feeds.

System Improvements

HUD Configuration Application

Hosted on ICS device, accessible via browser.

System Extensions

Allow users to create extensions for DashAR in Python, to offer expansive functionality (e.g. Third-party API integration)

11

Conclusions

Let's wrap this up.

Conclusions

DashAR is an attempt to realize the potential of Automotive HUDs.

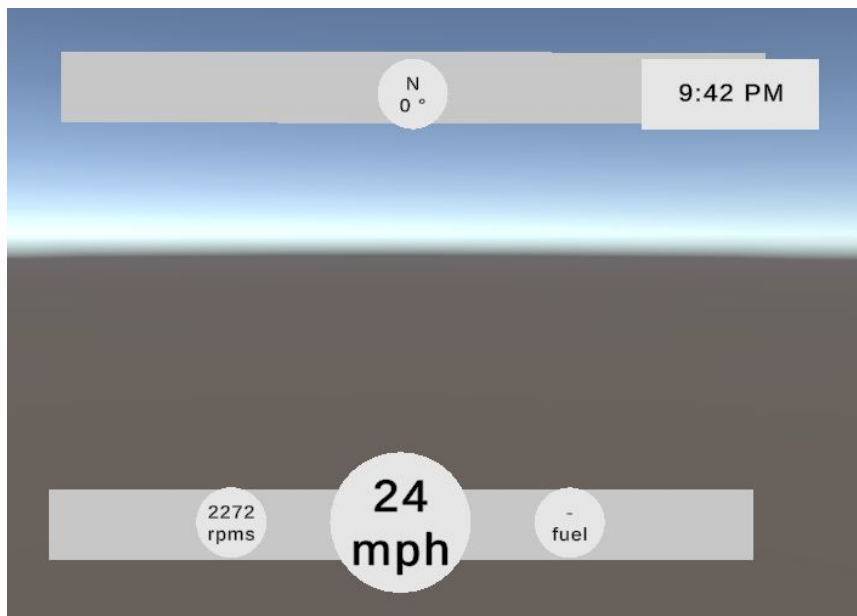
Automotive HUDs don't have to be fixed in design and in the information they provide

DashAR marries AR, HUDs, and Auto Instrumentation.

A convergence of three *mostly* disjointed technologies into a singular software system.

Acceptance through Open Source

Publish findings, provide reproducible, cost effective solution usable in any modern car, using any AR headset or data collection methods. An Open Source framework, extensible by free and non-free modules.



Thank you!

Do you have any questions?

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